

Original Expression

∑_{i=0}^{(a+c)^b} ⌊ i / (a+c) ⌋ - i mod (a+c)

Simplified Notation

Let $x = a + c$

Let $y = \frac{x^b}{x} = x^{b-1}$,

, so the expression becomes:

∑_{i=0}^{x^b} ⌊ i / x ⌋ - (i mod x)

This can be split into two separate sums: ∑_{i=0}^{x^b} ⌊ i / x ⌋ - ∑_{i=0}^{x^b} (i mod x)

Analysis of Each Component

First Sum: ∑_{i=0}^{x^b} ⌊ i / x ⌋

For $i = 0, 1, 2, \dots, x^b$:

- When $i = 0$: $\left\lfloor \frac{0}{x} \right\rfloor = 0$
- When $i = 1, 2, \dots, x - 1$: $\left\lfloor \frac{i}{x} \right\rfloor = 1$
- When $i = x, x + 1, \dots, 2x - 1$: $\left\lfloor \frac{i}{x} \right\rfloor = 2$
- And so on...

We notice that 1 repeats x times, 2 repeats x times, 3 repeats x times and so on

This gives us: ∑_{i=0}^{x^b} ⌊ i / x ⌋ = (1x + 2x + 3x + 4x...yx) + partial sum for remainder

In this problem $y = \frac{x^b}{x}$, so the remainder is 0

So we get the final form for the first sum:

∑_{i=0}^{x^b} ⌊ i / x ⌋ = x · (1 + 2 + 3 + 4...y) = xy · (y + 1) / 2

Second Sum: ∑_{i=0}^{x^b} (i mod x)

The modulo operation creates cycles of length x :

- For $i = 0, 1, 2, \dots, x - 1$: $(i \bmod x) = 0, 1, 2, \dots, x - 1$
- This pattern repeats every x terms, so in total it repeats $(\frac{x^b}{x} = y)$ times
- $(0 + 1 + 2 + 3 + \dots + x - 1) = \frac{x \cdot (x - 1)}{2}$, for y repetitions it becomes $y \cdot \frac{x \cdot (x - 1)}{2}$

So: ∑_{i=0}^{x^b} (i mod x) = xy · (x - 1) / 2

Final Expression

∑_{i=0}^{x^b} ⌊ i / x ⌋ - ∑_{i=0}^{x^b} (i mod x) = xy · (y + 1) / 2 - xy · (x - 1) / 2 = xy · ((y + 1) / 2 - (x - 1) / 2)

∑_{i=0}^{x^b} ⌊ i / x ⌋ - ∑_{i=0}^{x^b} (i mod x) = xy · ((y + 1 - x + 1) / 2) = xy · ((y - x + 2) / 2) = x^b · ((y - x + 2) / 2)